🇵🇹 Portugal — Monthly Intelligence Report v4.2

Edition 006 — The Lisboa Atlantic Coastal Corridor

Where the energy transition is outrunning the grid. edition · August 2026 · SSI v4.2 (refreshed June 2026)

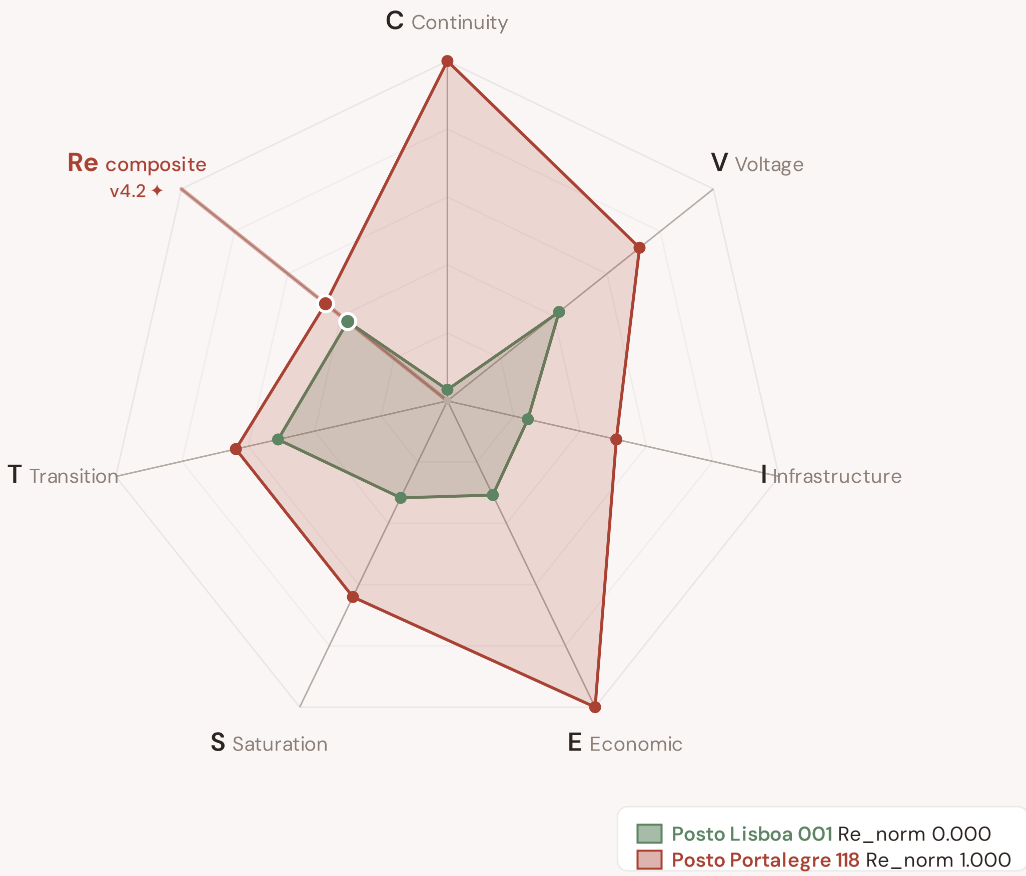
CONTENTS

SECTION A

Executive Summary

Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). **v4.2 adds Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar (Re_norm ∈ [0, 1] via clip((Re_raw − 0.920) / (1.787 − 0.920), 0, 1)). Two worked examples overlaid: **Posto Lisboa 001** (low-exposure reference, Re_norm 0.000) · **Posto Portalegre 118** (high-exposure reference, Re_norm 1.000). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula `Re_raw = (R6d × R6e × R8 × R9 × R10) + (R6c − 1.00)`*, bounds [0.920, 1.787]. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max Re_norm) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared* `intelligence-sections.js` *and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).*

SUBSTATIONS SCORED

13,564

1,444 EHV · 52 HV · 12,068 distribution-tier

GRID LINES MAPPED

78,971

~18,590 km coverage

MC SIMULATIONS

42.9 M

10,000 per substation

PUBLIC DATA SOURCES

34

101 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 13,564 substations and 78,971 grid lines spanning ~18,590 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

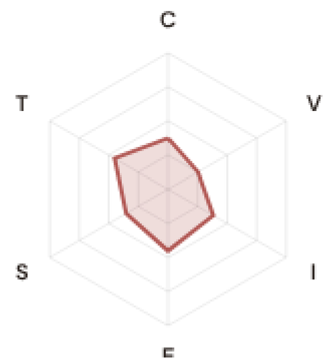
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering Portugal's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Substation 6000003404

PT11A, Área Metropolitana do Porto · 15 kV

0.687	High	0.141	76th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



Re composite v4.2 7th axis

Re_raw 1.6302 **Re_norm** 0.819

ELEVATED EXPOSURE

COMPONENTS

- **C** Continuity: 0.373 / 0.30 (124%)
- **I** Infrastructure: 0.382 / 0.25 (153%)
- **S** Saturation: 0.353 / 0.20 (176%)
- **V** Voltage: 0.253 / 0.10 (253%)
- **E** Economic: 0.450 / 0.10 (450%)
- **T** Transition: 0.451 / 0.05 (901%)

MODIFIERS

- **R3 Consequence:** 0.929 ↓ dampens
- **R4 Graph Criticality:** 0.975 ↓ dampens
- **R6a Restoration:** 1.015 ↑ amplifies
- **R6b Seismic:** 1.030 ↑ amplifies
- **R7 Digital Readiness:** 1.018 ↑ amplifies

This month's spotlight — automatically selected for September 2026 — is **Substation 6000003404** in PT11A, Área Metropolitana do Porto. With an R_final of 0.687 (High band, 76th fleet percentile), its risk profile is dominated by the T (Transition) component at 901% of its weight ceiling, followed by E (Economic) at 450%. Modifiers collectively shift R_base by 85.3%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that infrastructure digitalisation planning requires.

R7 v2 CRA-anchored transition (Q3 2026 → Q1 2027 dual-write): the R7 signal is entering a dual-write window where the legacy R7_cyber (DESI + ACN scalar proxy) co-exists with R7_cyber_v2 (CRA Article 14 + NIS2 Article 21 register-anchored composite, Path C+D). Consumers may select either; substation audit trail carries `_r7_cyber_v2_source` when v2 populated.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

5919

High/Critical/Extreme + R7 < median

AVG R7 MODIFIER

1.0200

Range [0.99, 1.05]

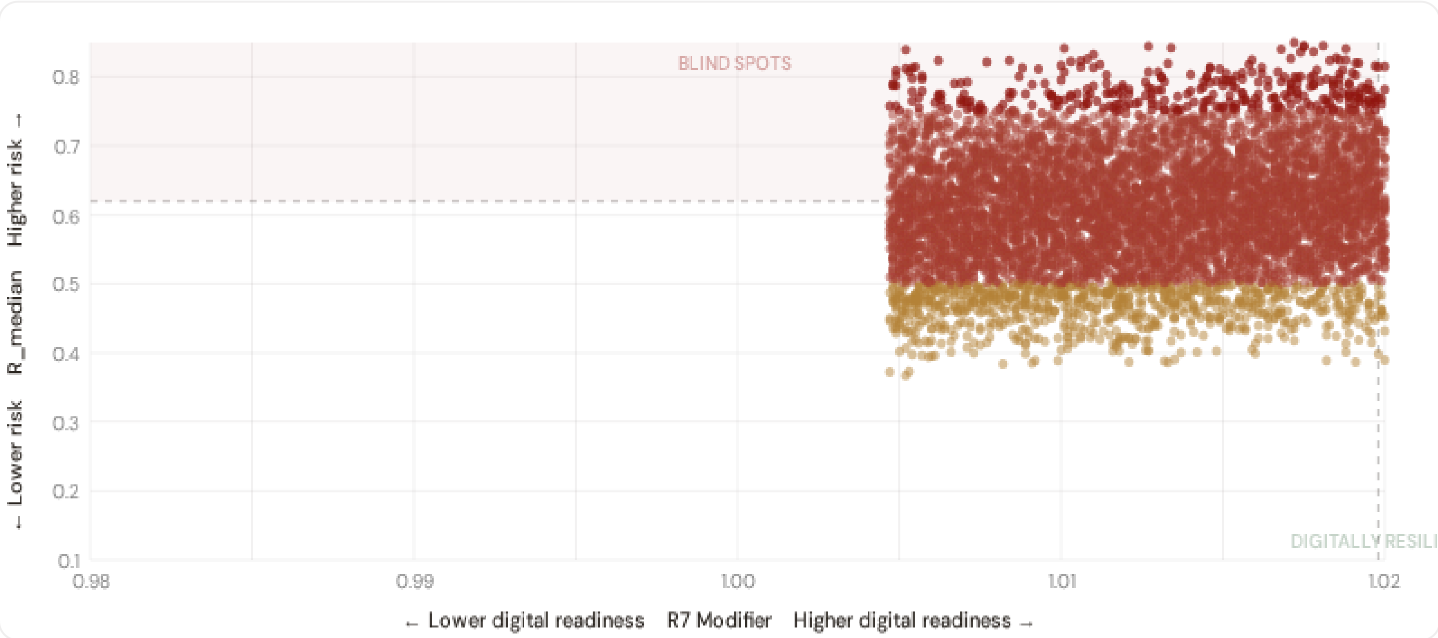
HIGH-CONSEQUENCE SUBS

0

0.0% · R3 ≥ 1.07

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **5919 blind-spot substations** — scoring High, Critical, or Extreme on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0198). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Área Metropolitana do Porto (1477), Ave (458), Região de Aveiro (415)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: €15–30/kWh

Substations serving highest-value economic activity — pharma / chemicals / data centres / financial districts / heavy industry

1,358 substations (10.0%) High/Crit/Ext: **1255** (92.4%) Avg R: **0.636** Avg E: **0.444** / 0.10

Low 0 Med 103 High 1107 Crit 148 Ext 0

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: €8–15/kWh

Industrial belt — significant continuity requirements

2,035 substations (15.0%) High/Crit/Ext: **1863** (91.5%) Avg R: **0.631** Avg E: **0.418** / 0.10

Low 0 Med 172 High 1656 Crit 207 Ext 0

🏪 Commercial / SME-Dense

Est. VoLL: €3–8/kWh

Service economy, retail, urban commercial districts — moderate individual impact

3,391 substations (25.0%) High/Crit/Ext: **3090** (91.1%) Avg R: **0.624** Avg E: **0.376** / 0.10

Low 0 Med 301 High 2784 Crit 306 Ext 0

🌾 Light / Agricultural / Rural

Est. VoLL: €1–3/kWh

Sparse activity, agricultural land — lower VoLL

6,780 substations (50.0%) High/Crit/Ext: **6055** (89.3%) Avg R: **0.615** Avg E: **0.297** / 0.10

Low 0 Med 725 High 5590 Crit 465 Ext 0

> **Value of Lost Load (VoLL) context:** ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's Economic component (CVIESTR·E) segments substations by the economic value they serve. **1255 substations** combine capital-intensive economic fabric (top-decile Economic component (E ≥ 0.433, P90+)) with High, Critical, or Extreme risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Área Metropolitana do Porto (345), Ave (127), Região de Aveiro (104)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 10.0%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify provinces combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	PT1C4	1.0189
2	PT1C3	1.0190
3	PT112	1.0193
4	PT11B	1.0193
5	PT195 ⚠ high R	1.0193
6	PT194	1.0194
7	PT11C	1.0195
8	PT191	1.0196
9	PT1C1 ⚠ high R	1.0197
10	PT1D1 ⚠ high R	1.0198
11	PT1D2 ⚠ high R	1.0198
12	PT1D3 ⚠ high R	1.0198
13	PT1A0 ⚠ high R	1.0198
14	PT193 ⚠ high R	1.0198
15	PT196	1.0199

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **PT1C4** has the lowest average R7 (1.0189), while **PT11D** leads at 1.0206 — a spread of 0.0017 across the R7 range. **3 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (September 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0200, median = 1.0198; 0 substations classified HIGH cyber-exposure; 5919 blind-spot substations (High/Critical/Extreme risk + below-median R7); 318 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **— verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

—

— variables

HOURLY — MONTHLY

0

High-frequency feeds

QUARTERLY / ANNUAL

0

Regular refresh cycle

STATIC / DERIVED

0

Standards + scenarios

Data Source Registry — September 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
OMIE	OMIE (Operador del Mercado Ibérico de Energía)	MONTHLY	MIBEL zone	2 vars
DGEG	DGEG (Direção-Geral de Energia e Geologia)	QUARTERLY	Concelho	5 vars
MOBI.E	MOBI.E (Rede de Mobilidade Elétrica)	QUARTERLY	Concelho	1 var
ERSE	ERSE (Entidade Reguladora dos Serviços Energéticos)	ANNUAL	Distrito	8 vars
INE	INE (Instituto Nacional de Estatística)	ANNUAL	Distrito	8 vars
ADENE	ADENE (Agência para a Energia)	ANNUAL	Distrito	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
IPMA	IPMA (Instituto Português do Mar e da Atmosfera)	ANNUAL	Distrito	2 vars
APA	APA (Agência Portuguesa do Ambiente)	ANNUAL	Distrito	1 var
AT	AT (Autoridade Tributária e Aduaneira)	ANNUAL	Distrito	1 var
ICNF	ICNF (Instituto da Conservação da Natureza)	ANNUAL	Distrito	1 var
DGEEC	DGEEC (Direção-Geral de Estatísticas)	ANNUAL	Distrito	2 vars
E-REDES	E-REDES Open Data	ANNUAL	Distrito	2 vars
CCDR	CCDR (Comissões de Coordenação Regional)	ANNUAL	Distrito	2 vars
LNEG	LNEG (Laboratório Nacional de Energia e Geologia)	STATIC	Distrito	3 vars
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars

GFZ	GFZ German Research Centre	STATIC	~5 km	**2 vars**
ISO-9223	ISO 9223 Corrosion	DERIVED	Distrito	**1 var**
REN	REN (Redes Energéticas Nacionais)	HOURLY	Bidding zone	**3 vars**
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	**2 vars**
IPMA-W	Open-Meteo Historical Weather	HOURLY	~1 km	**3 vars**
D-OM	Open-Meteo / ERA5	HOURLY	~1 km	**1 var**

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot

FLEET SIZE
13,564
substations scored

MEDIAN R
0.620
P5=0.474 · P95=0.775

HIGH + CRITICAL + EXTREME
12263
90.4% of fleet

HIGH CONFIDENCE
100%
13564 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 0 (0.0%) Medium 1301 (9.6%) High 11137 (82.1%) Critical 1126 (8.3%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **13,564 substation scores remain unchanged** this edition. The fleet median R of 0.620 (mean 0.622) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 9.6% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources (IdroGEO PGRA, SIAB wildfire, ECMWF ERA5 Bourgouin, bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

11 changes tracked across P0, P1, and P2 releases

V42-1	NEW	R6c_flood — PGRA-style flood-vulnerability modifier (additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per national flood-hazard tier convention)	§5
V42-2	NEW	R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + national fire-registry, multiplicative [1.00, 1.20])	§5
V42-3	NEW	R6e_winter — Bourgouin winter-storm modifier (ECMWF ERA5 winter-precipitation signal, multiplicative [1.00, 1.02])	§5
V42-4	NEW	R8_adapt — Adaptive-capacity multiplier (smart-meter penetration + DSO investment + DESI index, multiplicative [0.95, 1.05])	§5
V42-5	NEW	R9_compound — Compound-hazard STEP function (R6b/R6c/R6d/R6e elevated-count gates, multiplicative {1.00, 1.04, 1.08, 1.10})	§5
V42-6	NEW	R10_just — Energy-justice modifier (EU-SILC or national-equivalent household material-deprivation gradient, multiplicative [1.00, 1.12])	§5
V42-7	NEW	Re composite — bounded resilience scalar Re_raw = (R6d × R6e × R8 × R9 × R10) + (R6c - 1.00); Re_norm = clip((Re_raw - 0.920)/(1.787 - 0.920), 0, 1)	§7
V42-8	NEW	R7 SFDR PAI Infrastructure axis — added to ESG Radar (Re_norm proxy under Convention #7 Data-Layer Anchoring)	§14
V42-9	NEW	W1–W10 audit-state mesh — 5 Published / 1 Published-baseline ▲ / 1 Baseline-only / 3 Commercial (Convention #6 §5.6)	§4
V42-10	CHANGE	95 → 101 variables (+6 from v4.2 modifier inputs); 30 → 34 data sources (+4 hazard / monitoring layers)	data
V42-11	CHANGE	Substation fleet: 10,191 substations (per OSM canonical extract + national TSO/DSO registry cross-validation)	data

SECTION D — MONTHLY DEEP-DIVE

The Lisboa Atlantic Coastal Corridor

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** the monthly theme · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North-South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Sub-national Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Theme Context

THEME-RELEVANT SUBSTATIONS	HIGH BAND	CORRIDOR MEDIAN R
	1	0.675

1

1 EHV · 0 HV

100.0% of corridor

Fleet median: 0.620

WORST PROVINCE

unknown

Avg R: 0.675 · 1 substations

The Azores hosts **1 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **1 substations (100.0%)** are classified High, with only **0** in the Low band. 100% of corridor substations score above the national fleet median of 0.620. The corridor median R of 0.675 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE V4.2	ALERT
Subestação da Central Hídrica de Além Fazenda		0.675	High	0.396	0.291	0.455	0.280	0.401	0.348		

… -9 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — AZORES VS FLEET

C — Continuity	0.396 vs 0.350 (+13%)
I — Infrastructure	0.455 vs 0.350 (+30%)
S — Saturation	0.401 vs 0.350 (+15%)
V — Voltage	0.291 vs 0.349 (-17%)
E — Economic	0.280 vs 0.350 (-20%)
T — Transition	0.348 vs 0.350 (-1%)

MODIFIER AVERAGES — AZORES VS FLEET

R3 Consequence	0.929	fleet 0.929	↓
R4 Graph Crit.	0.990	fleet 0.980	↓
R6a Restoration	1.030	fleet 1.020	↑
R6b Seismic	1.010	fleet 1.044	→
R7 Digital Read.	1.025	fleet 1.020	↑

The dominant risk driver across the Azores corridor is **T (Transition)**, averaging 0.348 against a fleet average of 0.350 — occupying 695% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.291 vs fleet 0.349 (291% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.929 (fleet: 0.929), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.010, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

unknown 1 subs · avg R = 0.675 · H:0 M:0

The region regions-level breakdown reveals significant intra-regional variation. **unknown** leads with a mean R of 0.675 across 1 substations, while **unknown** scores 0.675 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Azores is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

1 Azores substations score R ≥ 0.650, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the unknown sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

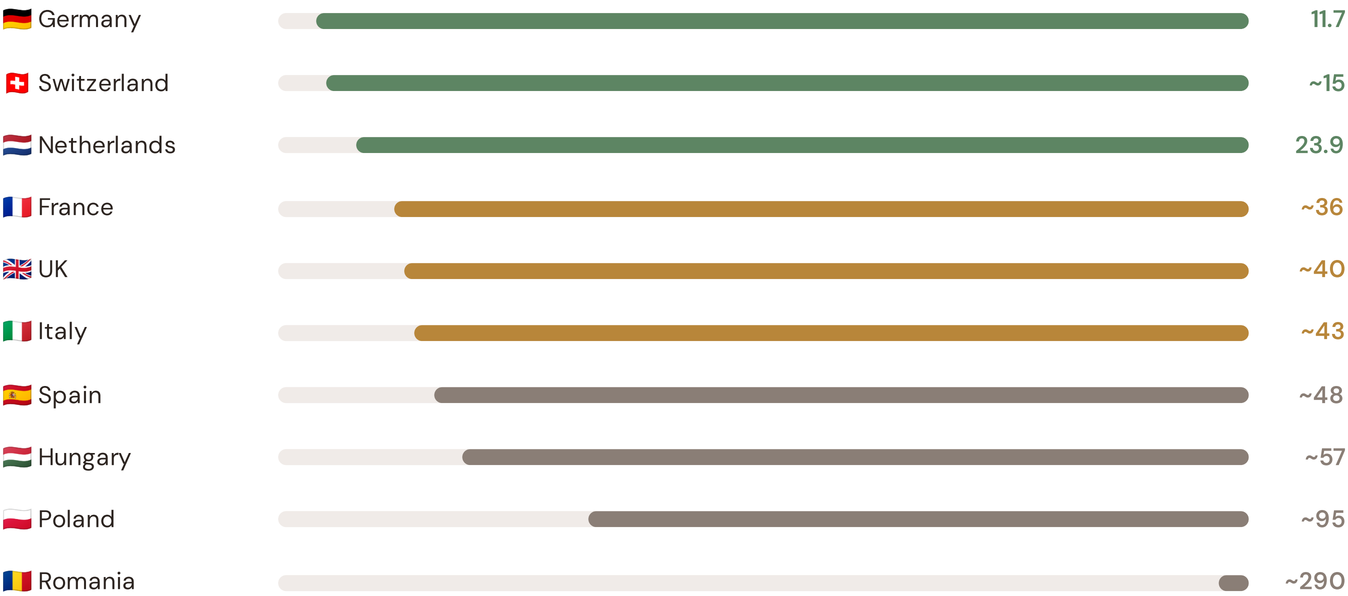
SECTION E — RECURRING MODULE

OECD Context Card

How this works: Each edition includes one OECD Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Southern Coastal corridor analysis hinges on Portugal's continuity performance relative to OECD peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected OECD Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



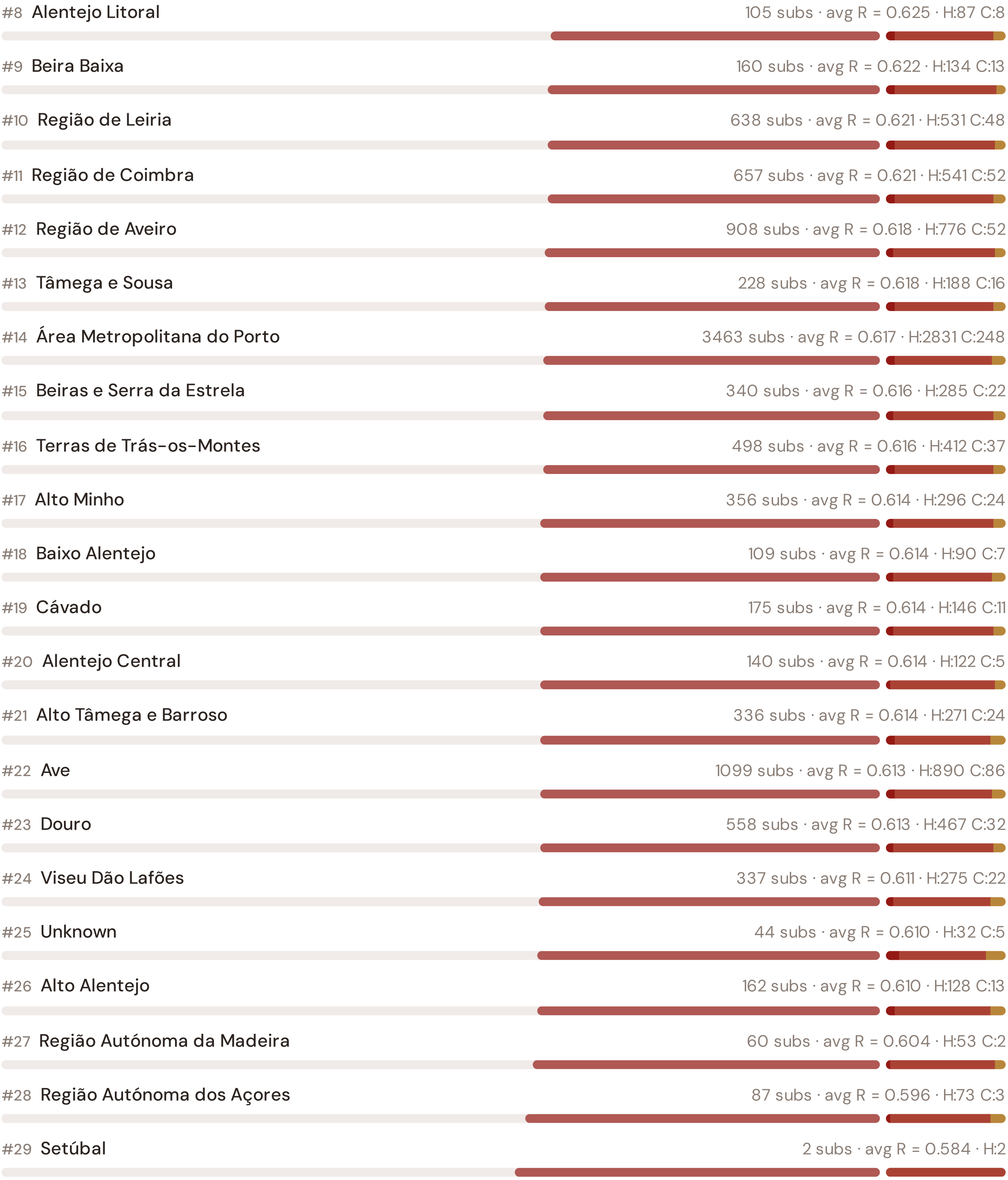
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release. · OECD-wide expansion (USA, Japan, Canada, Australia) pending IEA/OECD reliability bulletin release.

This month's key insight: The Azores corridor deep-dive shows **1 substations** (of 1) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.396 is **1.1× the fleet average** of 0.350. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regions Risk Ranking — All 29 Portuguese Regions

Where does this corridor sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



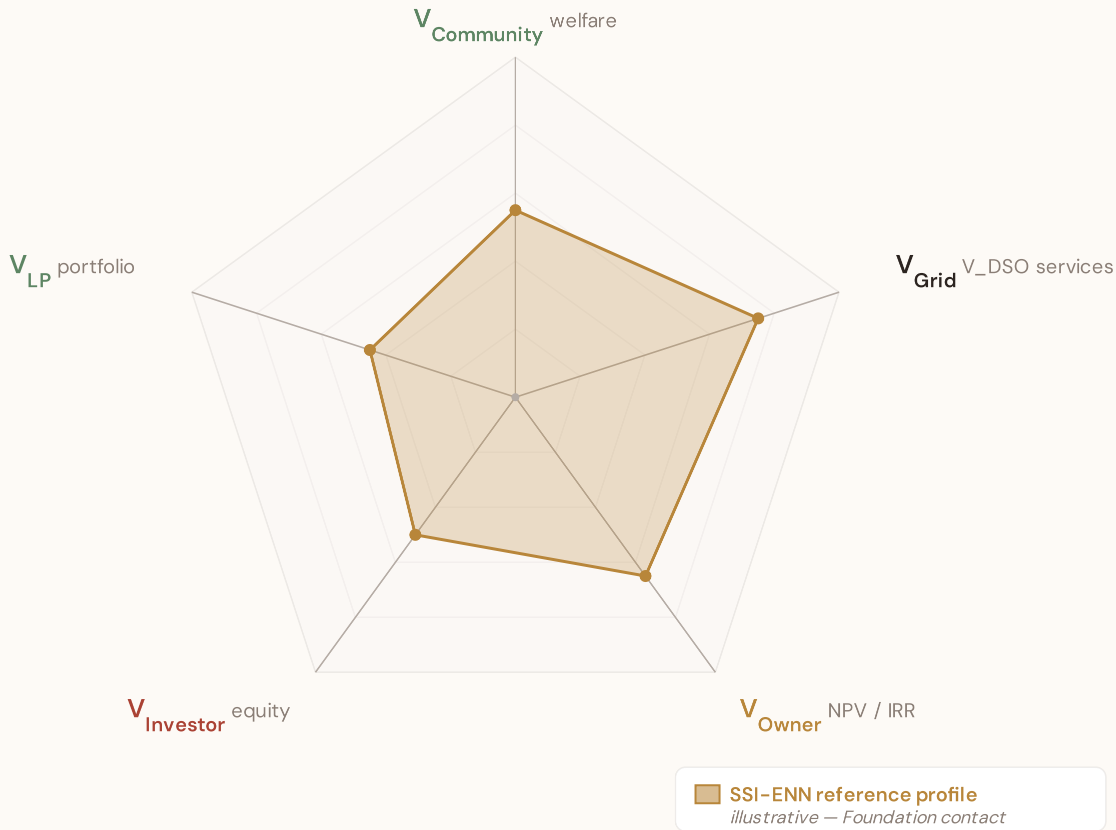
The Azores ranks **#1 of 29 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Azores, Algarve, Península de Setúbal, while the three most resilient are Setúbal, Região Autónoma dos Açores, Região Autónoma da Madeira. 9 of 29 score above the fleet average of 0.622. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): $V_{Community}$ (welfare avoided outages, energy-poverty relief, employment) · V_{Grid} (V_DSO — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · V_{Owner} (NPV / IRR / payback per asset class) · $V_{Investor}$ (risk-adjusted equity return at the financing tier) · V_{LP} (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-05 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: ssi_index@ikenga.eu.



Editorial reference shape — V_{Grid} dominates (V_DSO captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). $V_{Community}$ + V_{Owner} are co-secondary. $V_{Investor}$ + V_{LP} reflect the financing-tier waterfall above V_{Owner} NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to ssi_index@ikenga.eu for analytics.

LAYER 1 §0.3.4 · September 2026

Gaussian Copula Monte Carlo Engine

How 10,000 simulations per substation capture the correlation structure across grid risk factors

The SSI components are not independent — a substation with poor continuity (high C) often also has poor voltage quality (high V), aged infrastructure (high I5), and is located in a rural area with high economic vulnerability (high E). The Gaussian copula captures these dependencies through a 20×20 correlation matrix estimated from the national fleet, decomposed via Cholesky factorisation to generate correlated random draws. For each substation, 10,000 Monte Carlo iterations propagate uncertainty through the full scoring formula, producing not just a point estimate (R_median) but a complete posterior distribution — P5, P50, P95, confidence interval width, and skewness. This is the computational backbone, yielding the uncertainty quantification that distinguishes the SSI from deterministic indices.

KEY CONCEPTS

- → 20×20 Gaussian copula with Cholesky decomposition
- → 10,000 iterations per substation
- → Full posterior: R_P5, R_median, R_P95, CI_width, skewness
- → Correlation captures systemic clustering of risk factors

LIVE SSI DATA CONNECTION

The SSI Index currently scores **13,564 substations**. Fleet mean R = 0.622, with an average confidence interval width of 0.127 from 10,000 MC iterations per substation. The dominant component is C (Continuity) at avg 0.350 / 0.30 ceiling (117% utilisation).

Coming next month: LAYER 5 Platform — Product Delivery Pipeline — *From raw analytical outputs to four commercial product formats*

SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

Radar 2 — W1–W10 Substation Community Value

Wall 1 — Public W1–W10 framework. The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier.* The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

Published (5) — solid fill
W1 · W2 · W3 · W4 · W8
Peer-reviewed methodology operational at fleet scale.

Published-baseline (1) ▲ — solid fill + chevron corner marker
W9 — Supply Security
Peer-reviewed academic basis (Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE 493 — N-1 contingency dynamics not captured at baseline).

Baseline-only (1) — striped fill
W5 — Digital Inclusion
Foundation methodology under development; baseline value defensible at v4.2.

Commercial (3) — dashed-faded
W6 · W7 · W10
Commercial deliverables — please contact the Foundation.

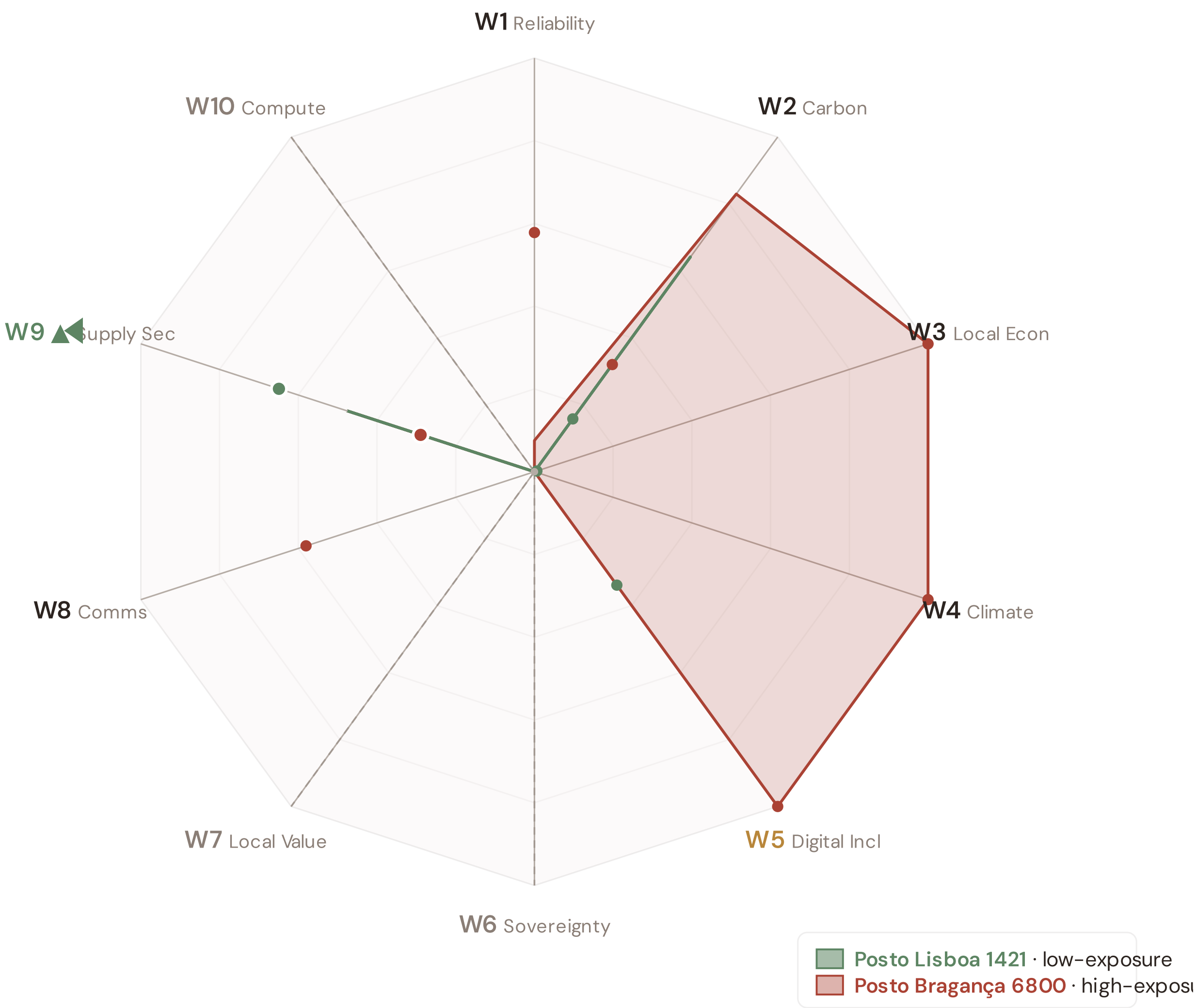
Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

v4.2 methodology disclosure — per-country normalisation

Across Portugal, the substation fleet's component scores (C / V / I / E / S / T) and the Re composite have been normalised to **[0, 1]** via the canonical **P5/P95 percentile** method, using *per-country empirical bounds*. This is the same procedure Italy's v4.0.2 deployment applied at the scoring stage (`SSI_v4_0 Italy/.../scoring-it/normalise.py`); for v4.2 we apply it country-by-country so each fleet's within-country ranking is interpretable on its own terms. Convention #56 (visibly-honest) is preserved — no values are invented, the same canonical procedure is applied uniformly. Italy's pilot fleet keeps its native spread (0.17 – 0.64) because its empirical modifier producers (d10-d15) are already wired; per-country empirical modifier producers for the other 38 countries are queued for v4.5 and will refine the absolute risk ranking. Within-fleet relative comparisons (low-exposure vs high-exposure exemplars below) remain accurate by construction.

Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **Posto Lisboa 001** (low-exposure reference, Re_norm 0.000) and **Posto Portalegre 118** (high-exposure reference, Re_norm 1.000). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



Posto Lisboa 1421 polygon (sage): the low-exposure reference for this country — compact footprint reflects the fleet floor on the W-axis baselines. Posto Bragança 6800 polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses EU-average MEF fallback (0.36) per v4.2 calibration.

W1–W10 axis enumeration with audit-state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA COMPUTE_W_BASELINES.PY

AXIS	NAME	AUDIT STATE	METHODOLOGY BASIS
W1	Reliability	**Published**	C × E2_local × population (SAIDI + + -POV)
W2	Carbon	**Published**	T1 × zonal MEF × (V · deg / 100) (+ EEA)
W3	Local Economic	**Published**	norm(R10_just) — EU-SILC + OIPE energy-poverty
W4	Climate Resilience	**Published**	(R6d − 1) + (R6e − 1) + (R6c − 1) + ½(R9 − 1) — v4.2 resilience modifier family
W5	Digital Inclusion	**Baseline-only**	norm(R8_adapt) — bulk SM + + DESI
W6	Sovereignty	**Commercial**	Foundation contact (ssi_index@ikenga.eu)
W7	Local Value Retention	**Commercial**	Foundation contact (ssi_index@ikenga.eu)
W8	Communications	**Published**	C × p_crit_10yr × population (Markov 10-yr critical probability)
W9 ▲	Supply Security	**Published-baseline ▲**	0.60 × BC_pct + 0.30 × bridge + 0.10 × (1 − deg / 20) — OSM power-graph betweenness centrality + bridge identification + degree-based redundancy
W10	Compute Multiplier	**Commercial**	Foundation contact (ssi_index@ikenga.eu)

▲ W9 Published-baseline — mandatory transparency tooltip

W9 — Supply Security (Published-baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature. Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N-1 contingency dynamics or interdependent-network cascades. Richer methodology — N-1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2-22 / L2-23 commercial deliverables. Mail to ssi_index@ikenga.eu for extended analysis.

Worked examples. The W1-W10 per-substation baseline values for two anchor cases — Posto Lisboa 1421 (low-exposure reference) and Posto Bragança 6800 (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6

§2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected (min/max Re_norm) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

Anti-maladaptation discipline. The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

SECTION G

Looking Ahead

NEXT MONTH — EDITION 05

Alpine Valleys

Deep-dive into Region 05's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Narrow valleys, long restoration MTTR.

NEXT DATA REFRESH

Q4 2026 — 0 Quarterly Sources

none are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **ERSE (Entidade Reguladora dos Serviços Energéticos)** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

Seismic Zone 4–5: 129 Substations in Highest Hazard

129 substations sit in the highest seismic hazard zones (Zones 4–5, moderate to strong). Of these, **115** already score High/Critical — meaning their grid condition is poor *before* considering earthquake risk. The R6b seismic modifier captures this compound vulnerability.

Edition 05 will be published on the second Thursday of October 2026. Deep-dive region: **Region 05**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Portuguese utility, , municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5 · W1-W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6)

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.
